

Jae Wook Jung

jung.jaew@northeastern.edu

• <https://www.linkedin.com/in/jae-wook-jung> • <https://github.com/Jae2Wook?tab=repositories>

EDUCATION

Northeastern University	PhD in Economics	Sep 2022 - Present
Brigham Young University-Idaho	Bachelor of Science in Economics & Mathematical Science Applied Statistics	Sep 2019 - Apr 2022
Myongji University	Associates of Electrical Engineering	Mar 2014-Dec 2017

EXPERIENCE

Northeastern University; Boston, MA

Economics Department

Research Assistant

May 2023- May 2025

- With Dr. Shantanu Khanna, Assistant Professor of Public Policy and Economics, Northeastern University
 - **The Effects of Artificial Intelligence on Jobs: Evidence from an AI Subsidy Program. Hellsten et al. (Work in progress).**
 - Participated in the writing of the literature review and methodology sections for study examining impact of Swedish AI subsidy program on firm-level employment and productivity outcomes
 - Implemented staggered synthetic difference-in-differences methodology to identify causal effects of AI adoption incentives, analyzing the underlying Stata code to ensure proper specification
- With Dr. Nithish Prakash, Professor of Public Policy and Economics, Northeastern University
 - **Shaping Future Success: Evidence from an Early Childhood Human Capital Formation Intervention. Saraswat et al. (Working paper)**
 - Conducted comprehensive data analysis for randomized controlled trial including data cleaning, summary statistics, balance tables, and heterogeneity analysis across treatment arms
 - Implemented specialized analytical techniques: sharpened False Discovery Rate correction for multiple hypothesis testing, nonparametric causal tree machine learning for heterogeneous treatment effects estimation, and attrition analysis for missing data patterns
 - **Shaping Minds: The Transformative Effects of Theater-Based Learning. Chaurey et al. (Working paper)**
 - Supervised two junior research assistants and participated in the writing of an article for *Ideas for India*
 - Constructed Anderson indices from n-point Likert scale survey data for regression analysis to address multiple hypothesis testing concerns

Teaching Assistant

Sep 2022- Apr 2024

- Graded exams and assignments, and assisted professors in class on Healthcare & Medical Econ, Health Econ & Health Policy, International Food Policy, and Economics of Sustainability

Brigham Young University-Idaho; Rexburg, ID

Economics Department and Mathematics Department

Curriculum Design Assistant

Jan 2022 - Apr 2022

- Reviewed and revised in-class materials for Probability Theory course, ensuring accuracy and clarity of mathematical content
- Developed supplementary educational materials and resources to enhance student comprehension of probability concepts

Teaching Assistant

Sep 2020- Dec 2021

- Tutored students in concepts of Intermediate Microeconomics, Principles of Micro/Macroeconomics, Quantitative Methods, and Differential Equations with Linear Algebra
- Graded exams and assignments, and assisted professors in class on Intermediate Microeconomics, Principles of Micro/Macroeconomics, Quantitative Methods, and Differential Equations with Linear Algebra

Research Associate - Brigham Young University-Idaho; Rexburg, ID

Sep 2021- Nov 2021

BYU-I Economics Students Consulting Group

- Conducted meetings with economic development officials across Idaho and Utah to assess regional business opportunities
- Researched and analyzed regional economic indicators including labor market conditions, wage structures, and commercial real estate costs to inform client recommendations

Research Associate - Research and Business Development Center; Rexburg, ID

Apr 2021- Jul 2021

Non-profit Organization

- Conducted market research and data analysis to map the structure and dynamics of the appliance OEM parts aftermarket industry
- Assessed market expansion opportunities and competitive risks to develop strategic recommendations for client market share growth using SWOT analysis

- Squad leader: Led a squad of 9 soldiers
- Trained and deployed military working dogs for specialized missions

Research

- **Minimum Wage Effects on Youth Education in a Dual Labor Market: Evidence from Thailand. Jae Wook Jung**
This paper examines the impact of Thailand's 300 Baht minimum wage policy, announced during the 2011 General Election, on educational choices among youth aged 15-17. Exploiting exogenous variation in minimum wage increases across provinces using a generalized difference-in-differences approach, I find no overall enrollment effects. However, the policy altered school-work composition by increasing full-time student enrollment and reducing not enrolled and working youth. Female students experienced a significant increase in enrollment, while male students showed no significant change. These divergent responses reflect dominated disemployment effects, with youth enrollment decisions shaped by observable differences in adult labor market outcomes by education level.
- **Competitive Federalism and Development: Evidence from India's Aspirational Districts Programme. Jung and Khanna**
In 2018, the Indian government launched the Aspirational Districts Programme to accelerate development in India's most underdeveloped districts. We examine the program's effects on labor markets, poverty, and structural change using a synthetic difference-in-differences approach on nationally representative data spanning 2009-2023. We find that the program significantly increased labor force participation rates, primarily driven by increases in rural areas and for women. The program led to increases in household consumption expenditures with associated reductions in poverty. We also find evidence of slowing urbanization in treated districts, accompanied by a shift towards self-employment and away from regular wage work, with gains concentrated in the primary sector.
- **Artificial Intelligence (AI) and Endogenous Productivity: Evidence from Korean Firms. Jung and Wang**
This paper studies how artificial intelligence (AI) adoption affects firm-level productivity in South Korea. Using a 2017–2023 panel covering all major market sectors, we combine direct survey measures of AI adoption with a structural model of endogenous productivity to address selection into adoption. We find AI adoption raises revenue-based productivity by about 5% on average, with substantial heterogeneity by time since adoption that is consistent with a delayed J-curve, by sector with the largest gains in ICT and short-run losses in manufacturing, and by application area with gains concentrated in product/service development and sales and marketing rather than production processes. Using the estimates, we also examine how AI adoption relates to market power.
- **Open to all? Diversity Statements in Job Advertisements and Hiring Discrimination. Granberg, Jung, Khanna, and Sinclair**
Diversity statements are increasingly common in job advertisements, particularly in developed countries. Using data from nearly 11,500 job advertisements in Sweden, we characterize the types of jobs and firms who choose to include diversity statements. Further, we examine how the presence of diversity statements interacts with ethnic and gender discrimination in hiring decisions. Jobs that include diversity statements are much more likely to come from smaller firms and occupations like chefs, haircare, and cleaners. Using a correspondence study design, we find evidence of lower callbacks for ethnic minorities and higher callbacks for women. Prima facie, discrimination levels appear lower for jobs with diversity statements relative to those without, but this is not the case once we account for differences in job and employer characteristics using entropy balancing methods. From the perspective of an ethnic minority worker, applying to jobs with diversity statements is not associated with higher callback rates or lower discrimination levels.

LEADERSHIP, ACTIVITIES, HONORS

Republic of Korea Army- Squad Leader

Dec 2016 - May 2017

- Led a squad to be in the best-prepared condition to perform the foreseen and unforeseen missions successfully. Eight soldiers under my command performed extremely well by following my leadership and example.

Jack & Mary Wheatley Leadership Scholarship

May 2020 - Apr 2022

- Administered for high academic excellence and leadership in the community

SKILLS

Computer Skills: Stata, Python, R, SQL, Matlab, Latex, MS Excel

Certifications: Certification of Electrical Technician (2016)